

JOINT PROBABILITY DISTRIBUTION

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Two Dimensional Discrete Random Variable

In many trials we study two characteristics of an outcome and we need a pair of numbers to describe the outcome. For example,

we may select a student at random from a group and measure his height (h) and weight (w).

We may select a worker from a factory and note his age(x) and salary (y).

We may select a couple at random and note the age of the husband (x) and the age of the wife (y).

We may select a student at random and note the marks in Mathematics (x) and marks in Statistics (y).

In such cases we need a pair of numbers to record the outcome. Such experiments are called **two dimensional**.

Two Dimensional Discrete Random Variable

- We consider two function which associate a pair of real numbers to an outcome of an experiment of this type. These two functions taken jointly are called a **two dimensional discrete random variable**.
- If X and Y assume finite or countable infinite values, then (X, Y) is called a two dimensional **discrete random variable**.

Two Dimensional Discrete Random Variable

Definition: Let (X, Y) be a two dimensional discrete random variable. With each possible outcome (x_i, y_j) we associate a number p_{ij} representing the probability of the event that $X = x_i$ and $Y = y_j$ and satisfying the conditions:

- (i) $p(x_i, y_j) \geq 0$ for all i, j
- (ii) $\sum_{j=1}^{\infty} \sum_{i=1}^{\infty} p(x_i, y_j) = 1$

The function p defined for all (x_i, y_j) is called the joint **probability mass function** of (X, Y) .

The set of values of (x_i, y_j) together with their probabilities p_{ij} is called the **probability distribution** of (X, Y)

Two Dimensional Discrete Random Variable

- Probability Distribution of (X, Y) is given by

X/Y	y_1	y_2			y_m
x_1	p_{11}	p_{12}			p_{1m}
x_2	p_{21}	p_{22}			p_{2m}
x_3	p_{31}	p_{32}			p_{3m}
x_n	p_{n1}	p_{n2}			p_{nm}

Marginal Probability Distribution

- A two dimensional probability distribution gives us the probability that X will take a particular value and Y will take a particular value.
- But we may also be interested in the probability that X will take a particular value irrespective of the values of Y .
- This is called **marginal probability distribution of X** .
- Or we may be interested in the probability that Y will take a particular value **irrespective of the values of X** .
- This is called **marginal probability distribution Y**
- Marginal probabilities distributions in the case of discrete two dimensional distributions are easily obtained by summing probabilities vertically or horizontally.

Marginal Probability Distribution

- In general, consider the following probability distribution of (X, Y) .

X/Y	y_1	y_2			y_m	Total
x_1	p_{11}	p_{12}			p_{1m}	p_1
x_2	p_{21}	p_{22}			p_{2m}	p_2
x_3	p_{31}	p_{32}			p_{3m}	p_3
x_n	p_{n1}	p_{n2}			p_{nm}	p_n
Total	p_1'	p_2'			p_m'	1

Marginal Probability Distribution

- If we find the probabilities $p_1, p_2, \dots, p_i, \dots, p_n$ of $x_1, x_2, \dots, x_i, \dots, x_n$ only, irrespective of the values taken by Y, the probabilities distribution so obtained is called the marginal probability distribution of X.

Thus, the marginal probability distribution of x is:

X	$x_1, x_2, \dots, x_i, \dots, x_n$	Sum
$P(X)$	$p_1, p_2, \dots, p_i, \dots, p_n$	1

Marginal Probability Distribution

- Similarly, if we find the probabilities $p'_1, p'_2, \dots, p'_j \dots \dots p'_m$ of $y_1, y_2, \dots, y_j \dots \dots y_m$ irrespective of the values taken by X, the probability distribution so obtained is called the marginal probability distribution of Y.
- Thus, the marginal probability distribution of Y is:

Y	$y_1, y_2, \dots, y_i, \dots, y_n$	Sum
$P(Y)$	$p'_1, p'_2, \dots, p'_i, \dots, p'_m$	1

Conditional Probability Distribution

- $$P(X = x_i | Y = y_i) = \frac{P(X=x_i, Y=y_i)}{P(Y=y_i)}$$

Example-1

The joint probability distribution of X and Y is given by

$$P(X = x, Y = y) = \frac{x+3y}{24}, x = 1,2; y = 1,2$$

Find the joint p.m.f s of X and Y , Also Find the Marginal Probability distributions of X and Y.

Solution:

X/Y	1	2	Total
1	$1/6$	$7/24$	$11/24$
2	$5/24$	$1/3$	$13/24$
Total	$3/8$	$5/8$	1

Example-1

- The Marginal Probability Distribution of X & Y is given by

X	1	2
P(X)	$\frac{11}{24}$	$\frac{13}{24}$

Y	1	2
P(Y)	$\frac{3}{8}$	$\frac{5}{8}$

Example-2

- The joint probability distribution of X_1 and X_2 is given by $P(X_1 = x_1, X_2 = x_2) = \frac{1}{27}(x_1 + 2x_2)$, $x_1 = 0, 1, 2$, $x_2 = 0, 1, 2$ Find the joint p.m.f s X_1 and X_2 , Also Find the Marginal Probability distributions of X_1 and X_2

Solution:

X_1/X_2	0	1	2	Total
0	0	$2/27$	$4/27$	$2/9$
1	$1/27$	$1/9$	$5/27$	$1/3$
2	$2/27$	$4/27$	$2/9$	$4/9$
Total	$1/9$	$1/3$	$5/9$	1

Example-2

- The Marginal Probability Distribution of X_1 & X_2 is given by

X_1	0	1	2
$P(X_1)$	$2/9$	$1/3$	$4/9$

X_2	0	1	2
$P(X_2)$	$1/9$	$1/3$	$5/9$

Example-3



3 balls are drawn at random without replacement from a box containing 2 white, 4 black and 3 red balls. If X denotes the number of white balls drawn and Y denotes the number of red balls drawn, find

- (i) Joint probability distribution of X and Y
- (ii) $P(X \leq 1)$,
 $P(X \leq 1, Y \leq 2), P(Y \leq 2 | X \leq 1), P(X + Y \leq 2)$
- (iii) Marginal Probability distribution of X
- (iv) Marginal Probability distribution of Y
- (v) Conditional distribution of X given $Y=1$

Also check whether X & Y are independent variables.

Example-3

$$(i) \quad P(X = 0, Y = 0) = P(\text{all 3 balls are black})$$

$$= \frac{{}^4C_3}{{}^9C_3} = \frac{1}{21}$$

$$(ii) \quad P(X = 0, Y = 1) = P(1 \text{ red ball \& 2 black balls})$$

$$= \frac{{}^3C_1 \times {}^4C_2}{{}^9C_3} = \frac{3}{14}$$

$$(iii) \quad P(X = 0, Y = 2) = P(2 \text{ red balls \& 1 black ball})$$

$$= \frac{{}^3C_2 \times {}^4C_1}{{}^9C_3} = \frac{1}{7}$$

$$(iv) \quad P(X = 0, Y = 3) = P(\text{all 3 balls are red})$$

$$= \frac{{}^3C_3}{{}^9C_3} = \frac{1}{84}$$

And so on ...

Example-3

The joint probability distribution of X and Y is given by

X/Y	0	1	2	3	total
0	$\frac{1}{21}$	$\frac{3}{14}$	$\frac{1}{7}$	$\frac{1}{84}$	$\frac{35}{84}$
1	$\frac{1}{7}$	$\frac{2}{7}$	$\frac{1}{14}$	0	$\frac{42}{84}$
2	$\frac{1}{21}$	$\frac{1}{28}$	0	0	$\frac{7}{84}$
total	$\frac{5}{21}$	$\frac{15}{28}$	$\frac{3}{14}$	$\frac{1}{84}$	1

Example-3

- (ii) $P(X \leq 1) = P(X = 0) + P(X = 1) = \frac{35}{84} + \frac{42}{84} = \frac{77}{84}$
- $P(X \leq 1, Y \leq 2)$
- $= P(X = 0, Y = 0) + P(X = 0, Y = 1) + P(X = 0, Y = 2) +$
 $P(X = 1, Y = 0) + P(X = 1, Y = 1) + P(X = 1, Y = 2)$
 $= \frac{1}{21} + \frac{3}{14} + \frac{1}{7} + \frac{1}{7} + \frac{2}{7} + \frac{1}{14} = \frac{19}{21}$

Example-3

$$P(Y \leq 2 | X \leq 1) = \frac{P(Y \leq 2, X \leq 1)}{P(X \leq 1)}$$

$$= \frac{19/21}{77/84} = \frac{76}{77}$$

$$P(X + Y \leq 2)$$

$$\begin{aligned} &= P(X = 0, Y = 0) + P(X = 0, Y = 1) + P(X = 0, Y = 2) \\ &\quad + P(X = 1, Y = 0) + P(X = 1, Y = 1) \\ &\quad + P(X = 2, Y = 0) \end{aligned}$$

$$= \frac{1}{21} + \frac{3}{14} + \frac{1}{7} + \frac{1}{7} + \frac{2}{7} + \frac{1}{21} = \frac{37}{42}$$

Example-3

- (iii) Marginal Probability distribution of X

X	0	1	2	Total
P(X)	35/84	42/84	7/84	1

Example-3

- (iv) Marginal Probability distribution of Y

Y	0	1	2	3	Total
P(Y)	$\frac{5}{21}$	$\frac{15}{28}$	$\frac{3}{14}$	$\frac{1}{84}$	1

Example-3

- (v) Conditional probability of X given Y=1
 $= P(X = 0|Y = 1) + P(X = 1|Y = 1) + P(X = 2|Y = 1)$

$$\begin{aligned} &= \frac{P(X = 0, Y = 1)}{P(Y = 1)} + \frac{P(X = 1, Y = 1)}{P(Y = 1)} + \frac{P(X = 2, Y = 1)}{P(Y = 1)} \\ &= \frac{3/14}{15/28} + \frac{2/7}{15/28} + \frac{1/28}{15/28} = 1 \end{aligned}$$

- (vi) Conditional probability of Y if X=2
 $= P(Y = 0|X = 2) + P(Y = 1|X = 2)$

$$\begin{aligned} &= \frac{P(Y = 0, X = 2)}{P(X = 2)} + \frac{P(Y = 1, X = 2)}{P(X = 2)} \\ &= \frac{1/21}{7/84} + \frac{1/28}{7/84} = 1 \end{aligned}$$

Example-3

- (vii) Check whether X & Y are independent variables
- X and Y are said to be independent if
- $P(X, Y) = P(X) \times P(Y) \quad \forall X \text{ \& } Y$

Here consider $X = 0, Y = 1$

$$P(X = 0, Y = 1) = \frac{3}{14}$$

$$\text{And } P(X = 0) \times P(Y = 1) = \frac{35}{84} \times \frac{15}{28} = \frac{25}{108}$$

Hence X & Y are not independent variables.

Two dimensional continuous Probability Distribution

Let (X, Y) be a two dimensional continuous variable and let $f_{XY}(x, y)$ be function of (x, y) such that

(i) $f_{XY}(x, y) \geq 0$,

(ii) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1$

(iii) $\int_a^b \int_c^d f_{XY}(x, y) dx dy$ represents the probability that $P(a \leq X < b, c \leq Y \leq d)$

then $f_{XY}(x, y)$ is called two dimensional probability density function.

Marginal Probability Distribution

(a) The marginal probability density function of X is obtained by integrating two dimensional p.d.f $f_{XY}(x, y)$ w.r.t y from $-\infty$ to ∞

Thus, the marginal p.d.f of X is $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$
where $f_X(x) \geq 0$ and $\int_{-\infty}^{\infty} f_X(x) dx = 1$

(b) The marginal probability density function of Y is obtained by integrating two dimensional p.d.f $f_{XY}(x, y)$ w.r.t x from $-\infty$ to ∞ .

Thus, the marginal p.d.f of Y is $f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$
where $f_Y(y) \geq 0$ and $\int_{-\infty}^{\infty} f_Y(y) dy = 1$

Example-4

The joint probability density of two random variables is

given by
$$f_{XY}(x, y) = \begin{cases} 15e^{-3x-5y}, & x > 0, y > 0 \\ 0, & \text{elseswhere} \end{cases}$$

(a) Find the probability that

(i) $1 < X < 2$ and $0.2 < Y < 0.3$ **(ii)** $X < 2$ and $Y > 0.2$

(b) Find marginal probability distributions of X and Y.

Example-4

$$(i) P(1 < x < 2, 0.2 < y < 0.3)$$

$$= \int_1^2 \int_{0.2}^{0.3} 15e^{-3x-5y} dx dy$$

$$= 15 \int_1^2 e^{-3x} dx \int_{0.2}^{0.3} e^{-5y} dy$$

$$= 0.00684$$

$$(ii) P(x < 2, y > 0.2)$$

$$= \int_0^2 \int_{0.2}^{\infty} 15e^{-3x-5y} dx dy$$

$$= 15 \int_0^2 e^{-3x} dx \int_{0.2}^{\infty} e^{-5y} dy$$

$$= 0.367$$

Example-4

(iii) Marginal Probability distribution of X

$$\begin{aligned}f_X(x) &= \int_0^{\infty} 15e^{-3x}(e^{-5y})dy \\&= -3e^{-3x}(0 - 1)\end{aligned}$$

$$\therefore f_X(x) = \begin{cases} 3e^{-3x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

Marginal Probability distribution of Y

$$\begin{aligned}f_Y(y) &= \int_0^{\infty} 15e^{-5y}(e^{-3x})dx \\&= -5e^{-5y}(0 - 1)\end{aligned}$$

$$\therefore f_Y(y) = \begin{cases} 5e^{-5y}, & y > 0 \\ 0, & \text{otherwise} \end{cases}$$

Example-5

Given $f_{xy}(x, y) = \begin{cases} cx(x - y), & 0 < x < 2, -x < y < x \\ 0, & \text{otherwise} \end{cases}$

(i) Evaluate c (ii) find $f_x(x)$ (iii) find $f_{\frac{y}{x}}\left(\frac{y}{x}\right)$

Solution: by the property of joint probability distribution

$$\int \int f_{xy}(x, y) dx dy = 1$$

$$\Rightarrow \int_0^2 \int_{-x}^x cx(x - y) dx dy = 1$$

$$\Rightarrow c = \frac{1}{8}$$

Example-5

$$(ii) f_x(x) = \int_{-x}^x f_{xy}(x, y) dy$$
$$= \int_{-x}^x \frac{1}{8} x(x - y) dy$$

$$= \frac{x^3}{4} \quad 0 < x < 2$$

$$(iii) f_{\frac{y}{x}}\left(\frac{y}{x}\right) = \text{conditional density function of } y \text{ given } x$$

$$= \frac{f_{xy}(x, y)}{f_x(x)}$$

$$= \frac{\frac{1}{8} x(x - y)}{\frac{x^3}{4}}$$

$$= \frac{x - y}{2x^2}, \quad -x < y < x$$

Example-6

$$f_{xy}(x, y) = xy^2 + \frac{x^2}{8}, 0 \leq x \leq 2, 0 \leq y \leq 1$$

Compute (i) $P(x > 1)$ (ii) $P\left(y < \frac{1}{2}\right)$ (iii) $P\left(x > 1, y < \frac{1}{2}\right)$

(iv) $P\left(y < \frac{1}{2} \mid x > 1\right)$ (v) $P(x < y)$ (vi) $P(x + y \leq 1)$

Solution:

$$\begin{aligned} \text{(i) } P(x > 1) &= \iint f_{xy}(x, y) dx dy \\ &= \int_{y=0}^1 \int_{x=1}^2 \left(xy^2 + \frac{x^2}{8} \right) dx dy \\ &= \frac{19}{24} \end{aligned}$$

Example-6

$$\begin{aligned} \text{(ii)} \quad P\left(y < \frac{1}{2}\right) &= \iint f_{xy}(x, y) \, dx \, dy \\ &= \int_{y=0}^{1/2} \int_{x=0}^2 \left(xy^2 + \frac{x^2}{8}\right) \, dx \, dy \\ &= \frac{1}{4} \end{aligned}$$

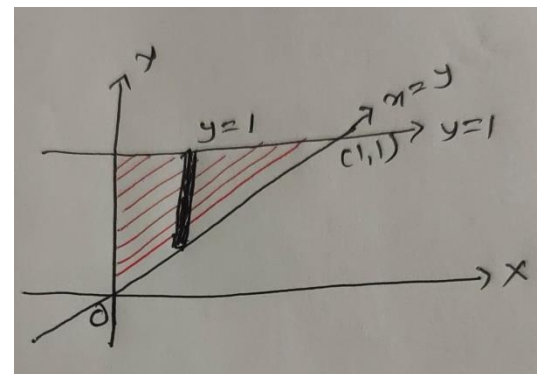
$$\begin{aligned} \text{(iii)} \quad P\left(x > 1, y < \frac{1}{2}\right) &= \int_{y=0}^{1/2} \int_{x=1}^2 \left(xy^2 + \frac{x^2}{8}\right) \, dx \, dy \\ &= \frac{5}{24} \end{aligned}$$

$$\text{(iv)} \quad P\left(y < \frac{1}{2} \mid x > 1\right) = \frac{P(x > 1, y < \frac{1}{2})}{P(x > 1)} = \frac{5/24}{19/24} = \frac{5}{19}$$

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Example-6

$$\begin{aligned} \text{(v)} \quad P(x < y) &= \int_{x=0}^1 \int_{y=x}^1 \left(xy^2 + \frac{x^2}{8} \right) dx dy \\ &= \frac{53}{480} \end{aligned}$$



$$\begin{aligned} \text{(vi)} \quad P(x + y \leq 1) &= \int_{x=0}^1 \int_{y=0}^{1-x} \left(xy^2 + \frac{x^2}{8} \right) dx dy \\ &= \frac{13}{480} \end{aligned}$$

